SEISMIC HAZARD ASSESSMENT FOR MAINLAND PORTUGAL

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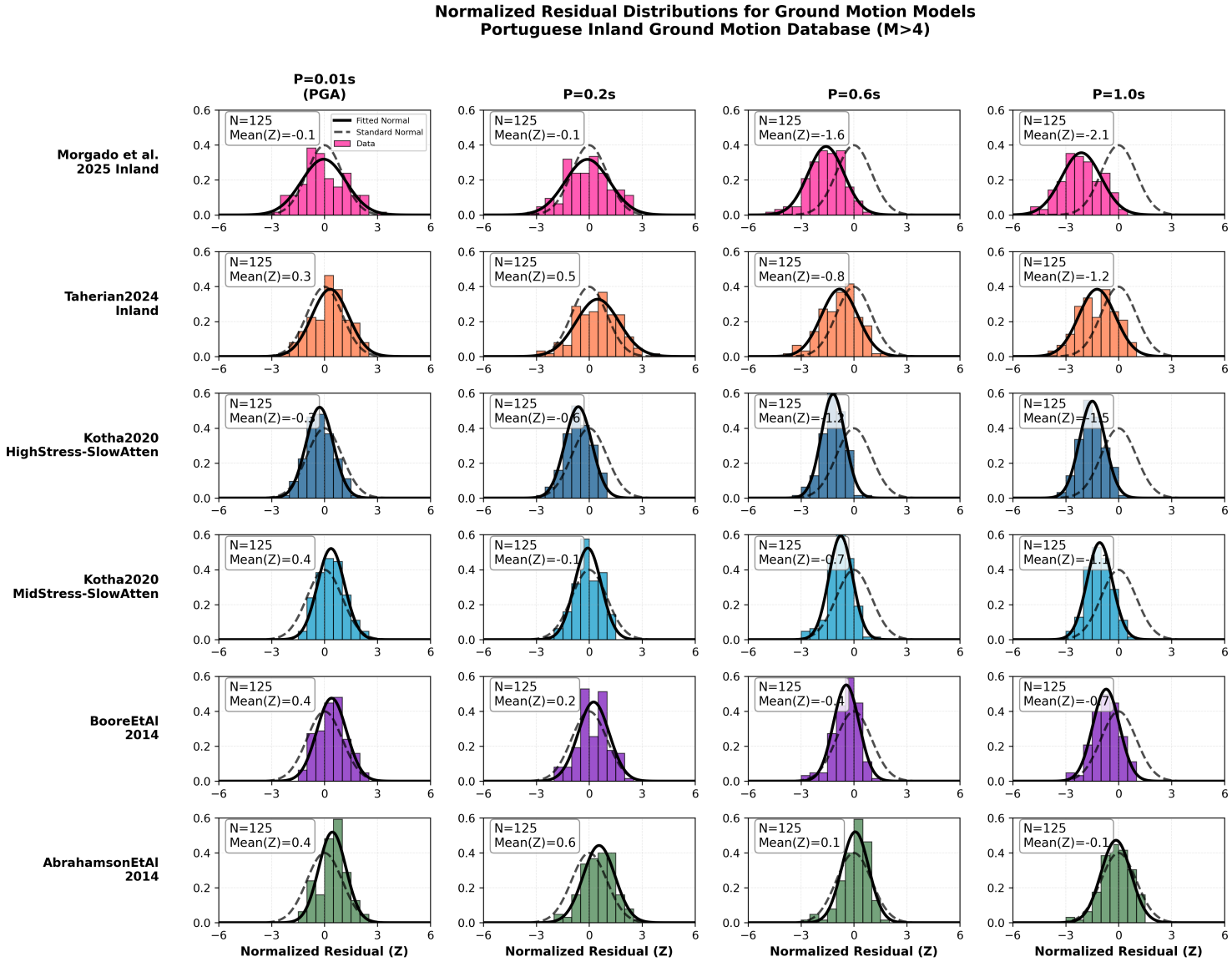
Past studies and current ground motion data indicate a large epistemic uncertainty in the selectio of ground motion models for Portugal. In particular, several studie indicate different seismogenic trend (e.g., stress drop, geometric attenuation) between inland and offshore events, due to the predominantly stable continental crust.

We performed residual anlysis considering a large suite of ground motion models to identify which proposal seem to predict ground shaking adequately. This figure presenta the results for 5 of the candidate models.



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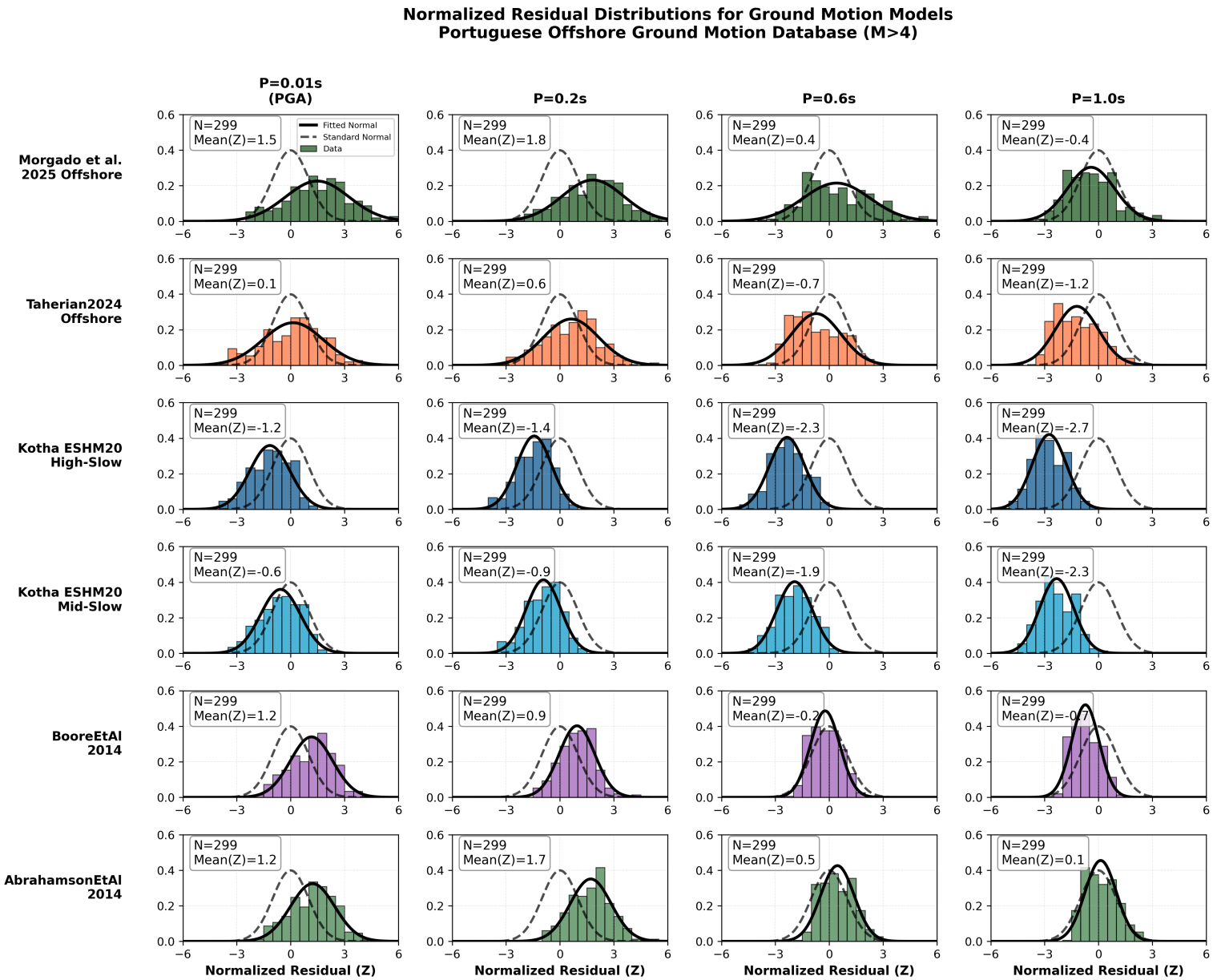

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The residual analyses allowed reducing the candidate models to 4 for inland and offshore events, as presented in this logic tree. We have selected 2 national models and 2 regional/global models for each type of event.

The following slides present the resulting seismic hazard (PGA for the 475 years return period) for mainland Portugal, considering different combinations of ground motion models.

National model

Regional/global model

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Seismic hazard (PGA for the 475 years return period) for mainland Portugal, using:

Inland events

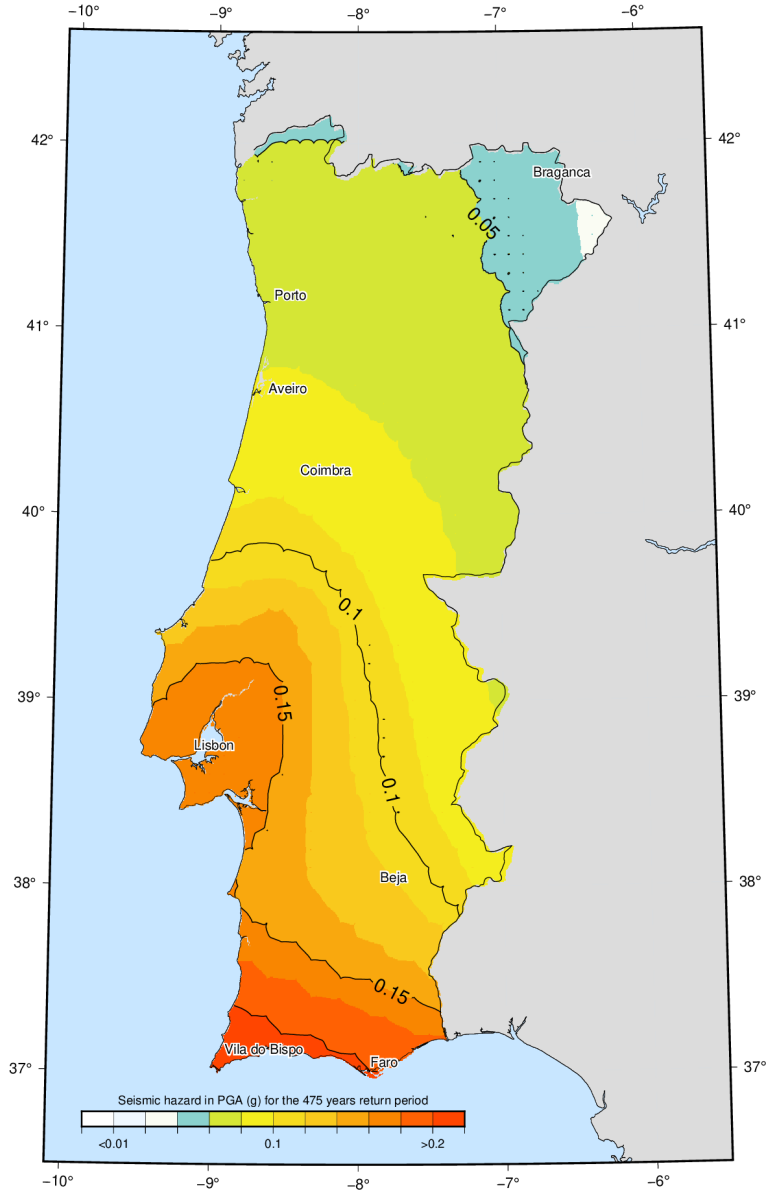
↓

Morgado et al. 2026 (inland)

Offshore events

↓

Morgado et al. 2026 (offshore)



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Seismic hazard (PGA for the 475 years return period) for mainland Portugal, using:

Inland events

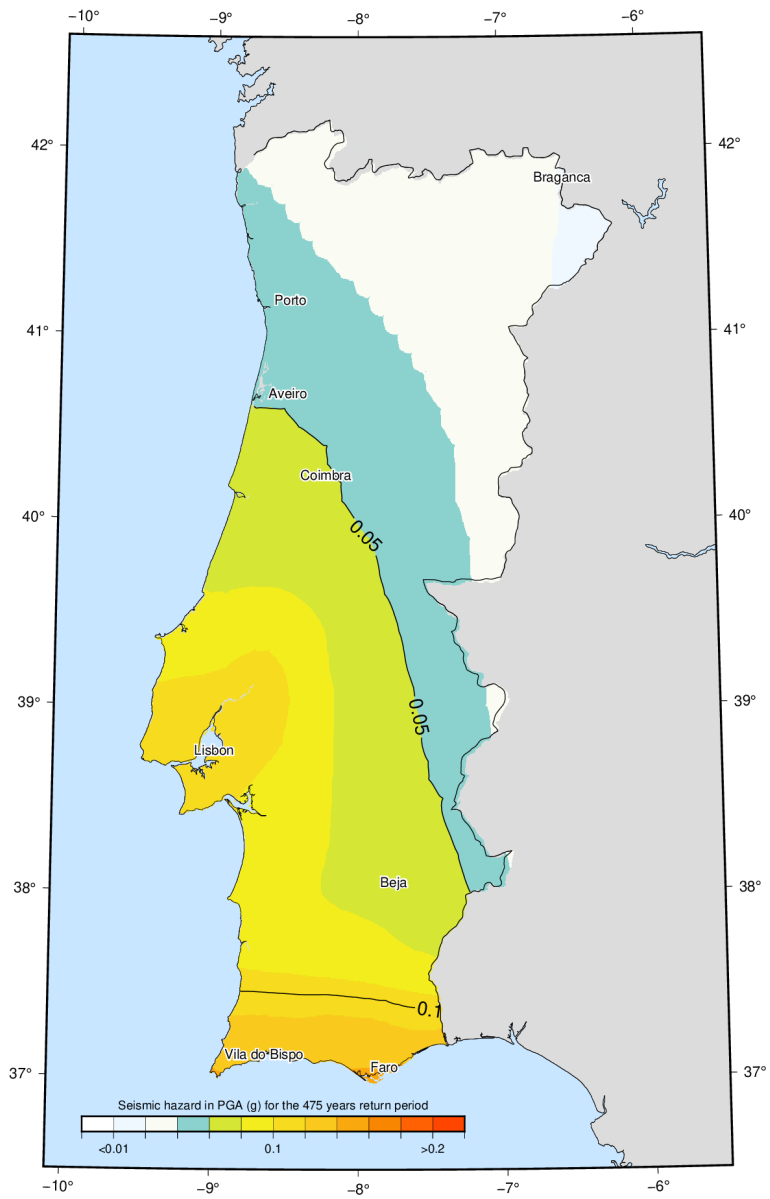
↓

Taherian et al. 2026 (inland)

Offshore events

↓

Taharian et al. 2026 (offshore)



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Seismic hazard (PGA for the 475 years return period) for mainland Portugal, using:

Inland events

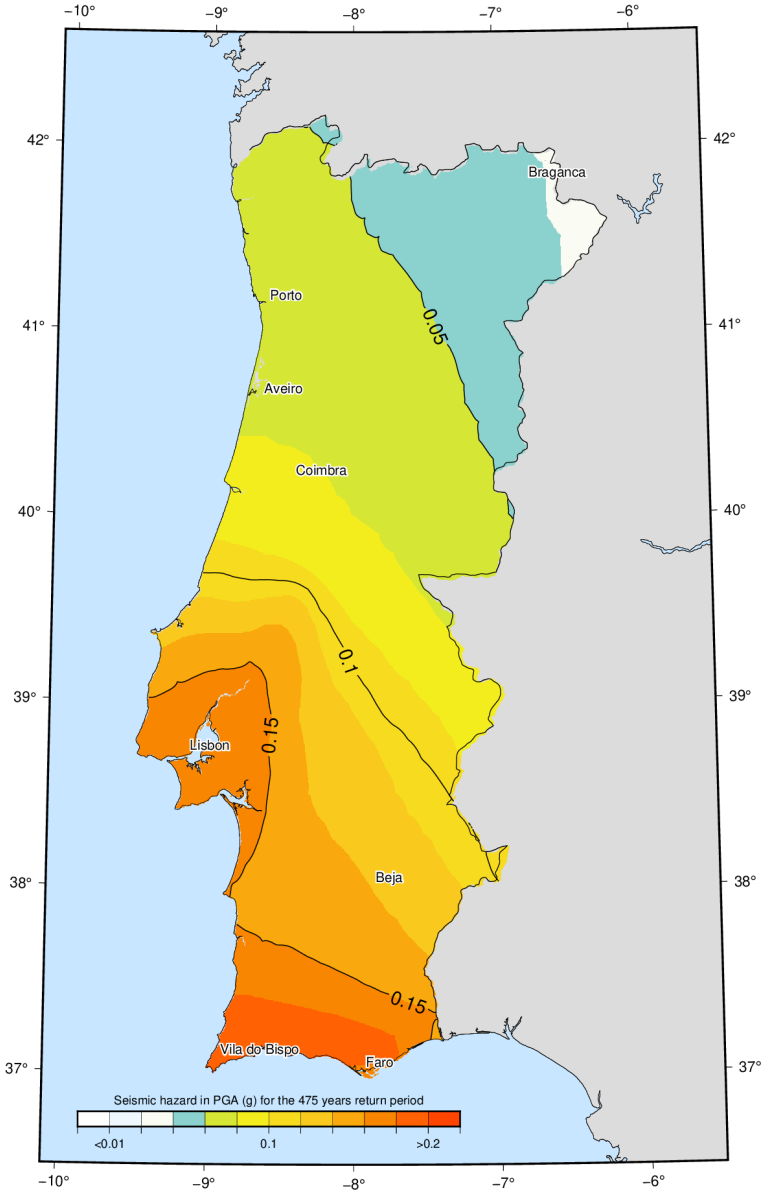
↓

Kotha et al. 2020 (mid-slow)

Offshore events

↓

Kotha et al. 2020 (high-slow)



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Seismic hazard (PGA for the 475 years return period) for mainland Portugal, using:

Inland events

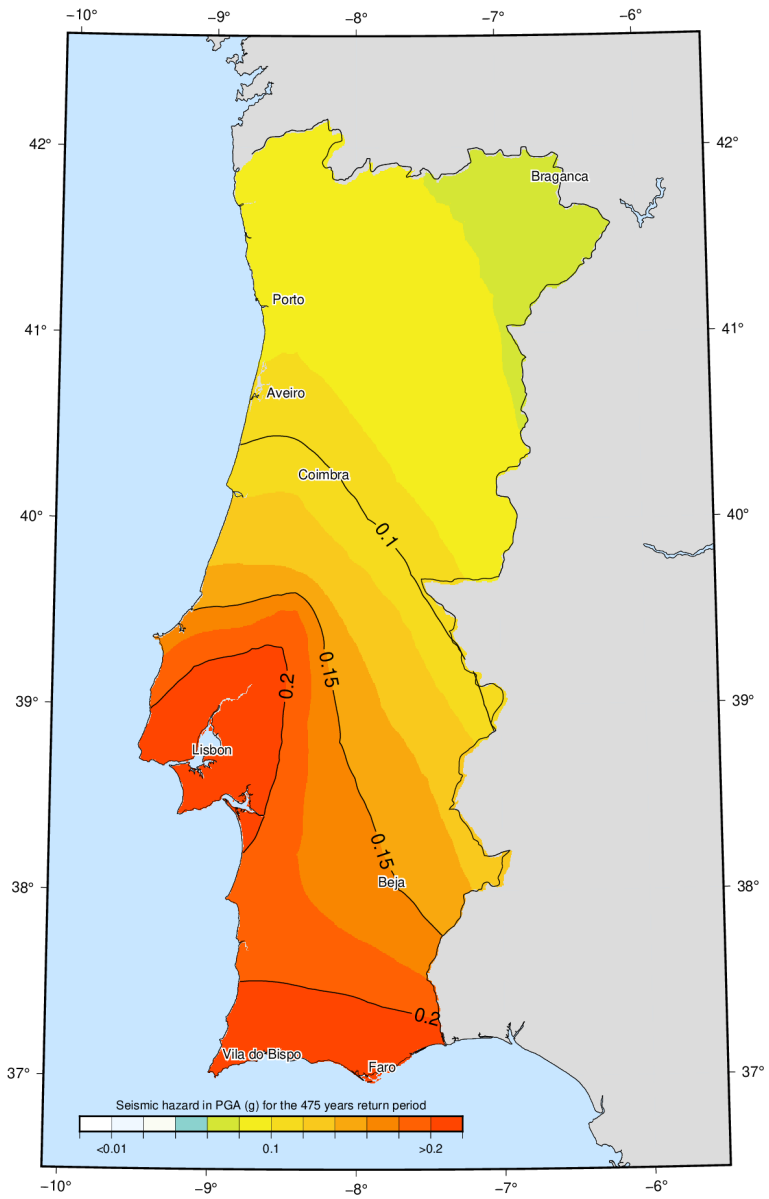
↓

Kotha et al. 2020 (high-slow)

Offshore events

↓

Kotha et al. 2020 (high-slow)



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Seismic hazard (PGA for the 475 years return period) for mainland Portugal, using:

Inland events

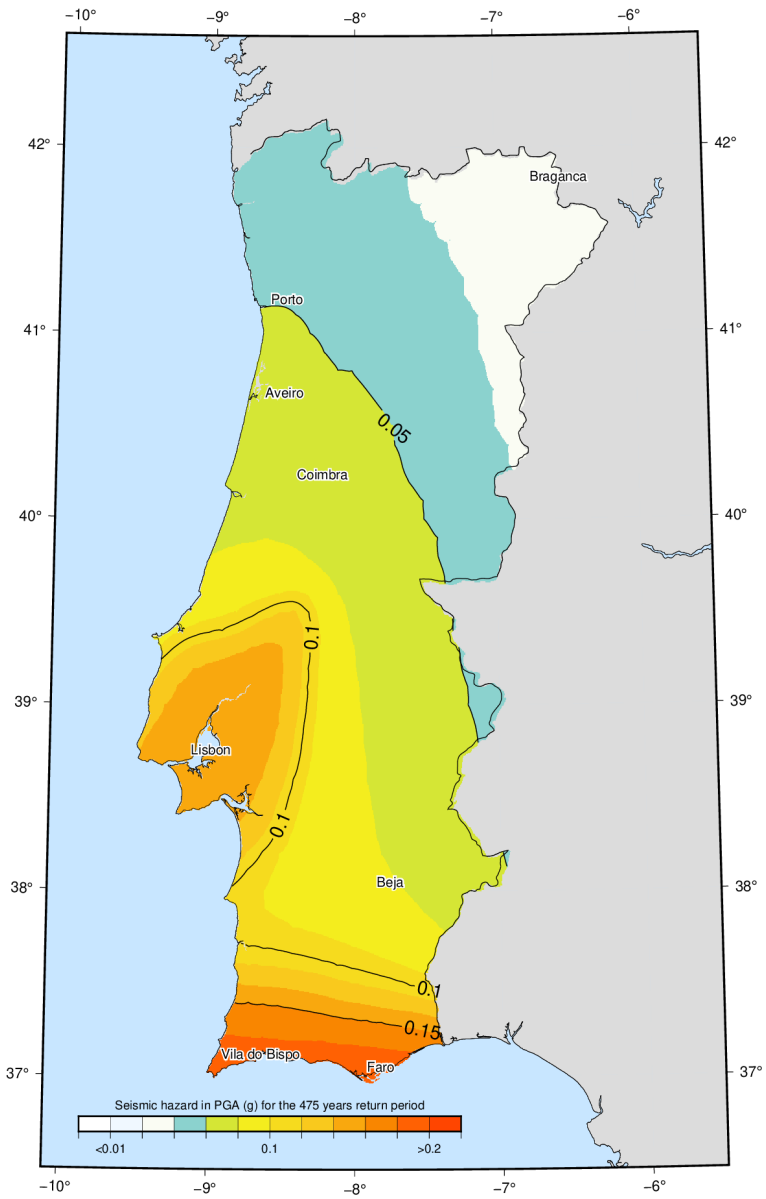
↓

Boore et al. 2014

Offshore events

↓

Abrahamson et al. 2014



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Next steps:

- Verify how the ground motion models estimate ground shaking for strong magnitudes
- Estimate seismic hazard disaggregation for specific locations in Portugal (Lisbon, Porto, Vila do Bispo)
- Investigate the influence of the weights from the various ground motion models
- Evaluate the potential consequences of changing the seismic demand in the safety and costs of the buildings